Enrolment Number										

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Winter, 2025

B. Tech 4th Semester Examination

Geotechnical Engineering– Theory

Course Code: BCE23403T

Full Marks – 50 **Time – 2 hours**

The figure in the margin indicates full marks for the questions.

SECTION 1

Choose the correct answer: **Marks: 10 x 1 =10**

1. (a) The soil transported by flowing water is called

(i) Aeolian soil (ii) Marine soil

(iii) Alluvial soil (iv) Lacustrine soil

(b) The degree of compaction of a soil is characterised by its

(i) consistency (ii) dry density

(iii) compressibility (iv) saturated unit weight

(c) A sample of soil has liquid limit 45%, plastic limit 25%, shrinkage limit 17% and natural moisture content 30%. The liquidity index of the soil is

(i) 5/20 (ii) 15/20

(iii) 8/20 (iv) 13/20

(d) A uniform sand stratum 2.5 m thick has a specific gravity of 2.62 and a natural void ratio of 0.62. The hydraulic head required to cause quick sand condition in the sand stratum is

(i) 2.32 m (ii) 2.50 m

(iii) 1.62 m (iv) 1.32 m

(e) Which of the following have an influence on the value of permeability?

(i) grain size (ii) void ratio

(iii) degree of saturation (iv) all of these

(f) The useful method of finding the shear strength of very plastic cohesive soil is by means of

(i) penetration test (ii) cone test

(iii) vane shear test (iv) torsional shear test

(g) A flow net constructed to determine the seepage through an earth dam having 4 flow channels and 16 equipotential drops. The coefficient of permeability in the horizontal and vertical direction are 4×10^{-7} m/s and 1×10^{-7} m/s respectively. If the storage head is 20 m, then the seepage per unit length of the dam in m^3/s will be

(i) 5×10^{-7} (ii) 40×10^{-7}

(iii) 20×10^{-7} (iv) 10×10^{-7}

(h) If the unconfined compressive strength of the soil is 50 kN/m^2, the cohesive strength of the soil is

(i) 100 kN/m^2 (ii) 25 kN/m^2

(iii) 50 kN/m^2 (iv) 75 kN/m^2

(i) A sample of saturated clay has a porosity of 0.562. The void ratio of the clay is

(i) 1.283 (ii) 0.438

(iii) 1.779 (iv) 0.360

(j) Quick sand is a

(i) moist sand containing small particles

(ii) condition which occurs in coarse sand

(iii) condition in which cohesionless soil loses its strength because of upward seepage flow.

(iv) none of the above.

SECTION 2

Answer any one question from this section **Marks: 10**

2(a) Define disturbed and undisturbed samples. The unit weight of a sand backfill was determined by field measurements to be 1746 kg/m^3. The water content at the time of test was 8.6 % and the unit weight of the solid constituents was 2.6 gm/cm^3. In the laboratory, the void ratios in the loosest and densest states were found to be 0.642 and 0.462 respectively. What was the relative density of the fill?

(4+6=10)

(b) An undisturbed sample of soil has a volume of 100 cm^3 and mass of 190 gm. On oven drying for 24 hours the mass is reduced to 160 gm. If the specific gravity of grains is 2.68, determine the water content, void ratio and degree of saturation of the soil. What are the different methods of surface exploration?

(6+4=10)

SECTION 3

Answer any one question from this section **Marks: 10**

3 (a) Explain the Casagrande's method of establishing the phreatic line of an earth dam. A homogenous anisotropic earth dam which is 20 m high is constructed on an impermeable foundation. The coefficient of permeability of soil used for the construction of the dam in the horizontal and vertical direction are 4.8×10^{-8} m/s and 1.6×10^{-8} m/s respectively. The water level on the reservoir side is 18 m from the base of the dam: downstream side is dry. It is seen that there are 4 flow channels and 18 equipotential drops in a square flow net drawn in the transformed dam section. Determine the quantity of seepage per unit length in m^3/s through the dam.

(5+5=10)

(b) What are the different factors affecting the permeability of soil? A horizontal stratified soil deposit consists of 3 uniform layers of thickness 7, 5 and 12 m respectively. The permeability of these layers are 7×10^{-4} cm/s, 5×10^{-4} cm/s and 12×10^{-4} cm/s, find the ratio of the effective average permeability of the deposit in the horizontal and vertical direction.

(5+5=10)

SECTION 4

Answer any one question from this section **Marks: 10**

4 (a) A silty clay layer 5 m thick were tested in a consolidometer and the following results were obtained- Initial void ratio = 0.90, Preconsolidation stress = 120 kN/m^2, Recompression index = 0.03, Compression index = 0.27. Evaluate the consolidation settlement if the present average overburden stress of the layer is 70 kN/m^2 and the increase in average stress in the layer is 80 k N/m^2.

(10)

(b) The following are the results of a compaction test

Mass of mould + wet soil (gm)	2925	3095	3150	3125	3070
Water content (%)	10	12	14.3	16.1	18.2

Volume of the mould is 1000 ml and mass of mould is 1000 gm. Take specific gravity of soilds = 2.70.

(i) Find the compaction curve showing the optimum moisture content and maximum dry density

(ii) Plot the zero air void line

(iii) Find the degree of saturation at the maximum dry density.

SECTION 5

Answer any one question from this section **Marks: 10**

5 (a) Determine the intensity of vertical stress and horizontal shear stress at a point 5 m below and 2 m horizontally away from the axis of loading of a 50 kN point load acting at the horizontal ground surface using Boussinesq and Westergaard's analysis

(10)

(b) A standard specimen of cohesionless sand was tested in triaxial compression and the sample failed at a deviator stress of 482 kN/m^2, when the cell pressure was 100 kN/m^2, under drained condition. Find the effective angle of shearing resistance of sand. Determine the deviator stress and the major principal stress at failure for another identical specimen of sand if it is tested under a cell pressure of 200 kN/m^2.

(10)